WORKED SOLUTION

STEP I 1992, Question 15

Probability and Statistics • Expected Waiting Time

STEP I 1992, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q15

The key idea

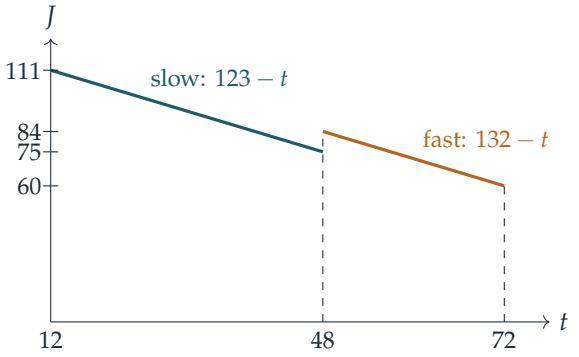
Everything repeats every hour, so it is enough to look at a passenger arriving at a time uniformly distributed over one hour. That hour splits into two gaps: the passengers in the gap before a train all catch that train, and each one's journey is "wait plus travel time". Within a gap the wait is uniform, so its average is half the gap. This turns the whole question into a quadratic in the length of one gap.

Solution

The present timetable

Measure time t in minutes past noon (any hour will do), and let the passenger's arrival time T be uniform on the hour-long interval $(12, 72]$. The trains in this window are the slow train at 48, arriving in London at $48 + 75 = 123$, and the fast train at 72 (that is, 12 minutes past the next hour), arriving at $72 + 60 = 132$.

- If $12 < T \leq 48$, the first available train is the slow one, so the journey time is $J = 123 - T$.
- If $48 < T \leq 72$, the first available train is the fast one, so $J = 132 - T$.



Since T has density $\frac{1}{60}$ on $(12, 72]$,

$$\mathrm{E}(J) = \frac{1}{60}\left[\int_{12}^{48}(123-t)\,\mathrm{d}t + \int_{48}^{72}(132-t)\,\mathrm{d}t\right].$$

Each integrand is linear, so each integral is the length of the interval times the value at its midpoint:

$$\int_{12}^{48}(123-t)\,\mathrm{d}t = 36\times(123-30) = 3348, \qquad \int_{48}^{72}(132-t)\,\mathrm{d}t = 24\times(132-60) = 1728.$$

Hence

$$\mathrm{E}(J) = \frac{3348+1728}{60} = \frac{5076}{60} = 84.6 \text{ minutes},$$

as required.

Retiming the fast train

Now let the fast train leave at x minutes past the hour, the slow one still at 48. Let d be the number of minutes from a fast departure to the next slow departure, so $0 \le d < 60$ and $d \equiv 48 - x \pmod{60}$. (For the original timetable, $d = 36$.) The hour then splits into

- a gap of length d before each slow train: those passengers wait W uniform on $[0, d)$ and then travel for 75 minutes;
- a gap of length $60 - d$ before each fast train: those passengers wait uniformly on $[0, 60-d)$ and then travel for 60 minutes.

A passenger lands in the first gap with probability $\frac{d}{60}$ and in the second with probability $\frac{60-d}{60}$, and a uniform wait on a gap of length g has mean $\frac{g}{2}$. So, by conditioning on which gap the passenger arrives in,

$$\begin{aligned}
60\,\mathrm{E}(J) &= d\left(\frac{d}{2}+75\right) + (60-d)\left(\frac{60-d}{2}+60\right)\\
&= \frac{d^2}{2} + 75d + \frac{3600-120d+d^2}{2} + 3600 - 60d\\
&= d^2 - 45d + 5400.
\end{aligned}$$

As a check, $d = 36$ gives $1296 - 1620 + 5400 = 5076$, matching the first part.

Ask yourself Why is the best gap not $d = 30$, splitting the hour evenly?

Each passenger in the slow train's gap pays a 15-minute penalty for its slower journey. So we want fewer passengers in that gap, which means making it shorter than half an hour, but not so short that the fast train's gap (and hence those passengers' waits) becomes long. The quadratic balances the two effects.

Completing the square,

$$d^2 - 45d + 5400 = \left(d - \tfrac{45}{2}\right)^2 + 5400 - \tfrac{2025}{4},$$

which is least when $d = 22.5$, a value inside $[0, 60)$. Then $60\,\mathrm{E}(J) = \frac{19575}{4}$, so the minimum average journey time is

$$\mathrm{E}(J) = \frac{19575}{240} = \frac{1305}{16} = 81\tfrac{9}{16} \approx 81.56 \text{ minutes}.$$

The fast train must leave 22.5 minutes before the slow one, so

$$x = 48 - 22.5 = 25.5,$$

that is, the fast train should leave at $25\frac{1}{2}$ minutes past the hour. If departures must be on whole minutes, $x = 25$ and $x = 26$ ($d = 23$ and $d = 22$) tie, each giving $\frac{2447}{30} \approx 81.57$ minutes.

Checks

- At $d = 0$ the two trains leave together; everyone takes the fast one and waits on average 30 minutes, so $\mathrm{E}(J) = 90$. The formula gives $\frac{5400}{60} = 90$.
- As $d \to 60$ the slow train leaves just before the fast one and catches nearly everyone: average wait 30, journey 75, total 105. The formula gives $\frac{3600 - 2700 + 5400}{60} = 105$.
- A numerical integration of the journey time over the hour for every x on a grid of step 0.1 gives 84.6 at $x = 12$ and a minimum of 81.5625 at $x = 25.5$.

Common slips

- Taking the passengers who arrive between 48 and 60 past the hour to the fast train "at 12" without adding the hour: their train leaves at 12 past the *next* hour.
- Averaging the two journey times weighted by the number of trains rather than by the lengths of the gaps; it is the length of the gap before a train that decides how many passengers it carries.

Beyond STEP

The quantity computed here is a mean waiting time for a periodic timetable, and the observation that long gaps catch more passengers is a form of the *inspection paradox*: a randomly arriving passenger is more likely to land in a long gap than a short one. This is a central idea in renewal theory and queueing theory, which study random arrivals and services. Grimmett and Welsh, *Probability: An Introduction*, is a good place to meet the underlying tools of conditioning and expectation.

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